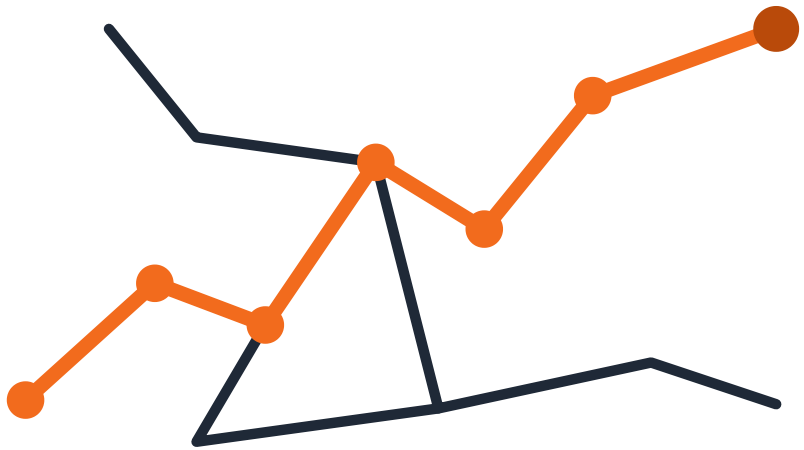
ADS Foundation workshop

Highway Accident Analysis in India | 2018-2022



Author:ADARSH AHUJA

Source: MoRTH / PIB

Made by Adarsh

QUESTIONS

Three questions frame the analysis

Source: MoRTH / PIB

01 Understand the reported National Highway accident trend from 2018 to 2022

02 Identify states with higher reported National Highway accident counts

03 Compare the National Highway share with all-road totals

Interpretation stays descriptive: aggregated annual reporting, not individual crash data.

Made by Adarsh

DATASET

The dataset combines annual totals with a focused highway view

Source: MoRTH / PIB

MoRTH Road Accidents in India 2022

Annual state/UT all-road accidents, fatalities, and injuries for 2018-2022

India-wide National Highway accident counts for 2018-2022

2022 top-10 states' National Highway accident counts for 2018-2022

National Highway fatalities for 2022

Scope note: the state-level National Highway series covers the report's top ten only, not all states.

Made by Adarsh

WORKFLOW

From raw CSV to a Power BI dashboard

01	02	03	04	05
Raw CSV	Python Pandas/NumPy cleaning and validation	SQL tables, joins, GROUP BY, CASE, and KPIs	Power BI model and DAX	Dashboard

Handle missing values explicitly

Check duplicate state-year keys

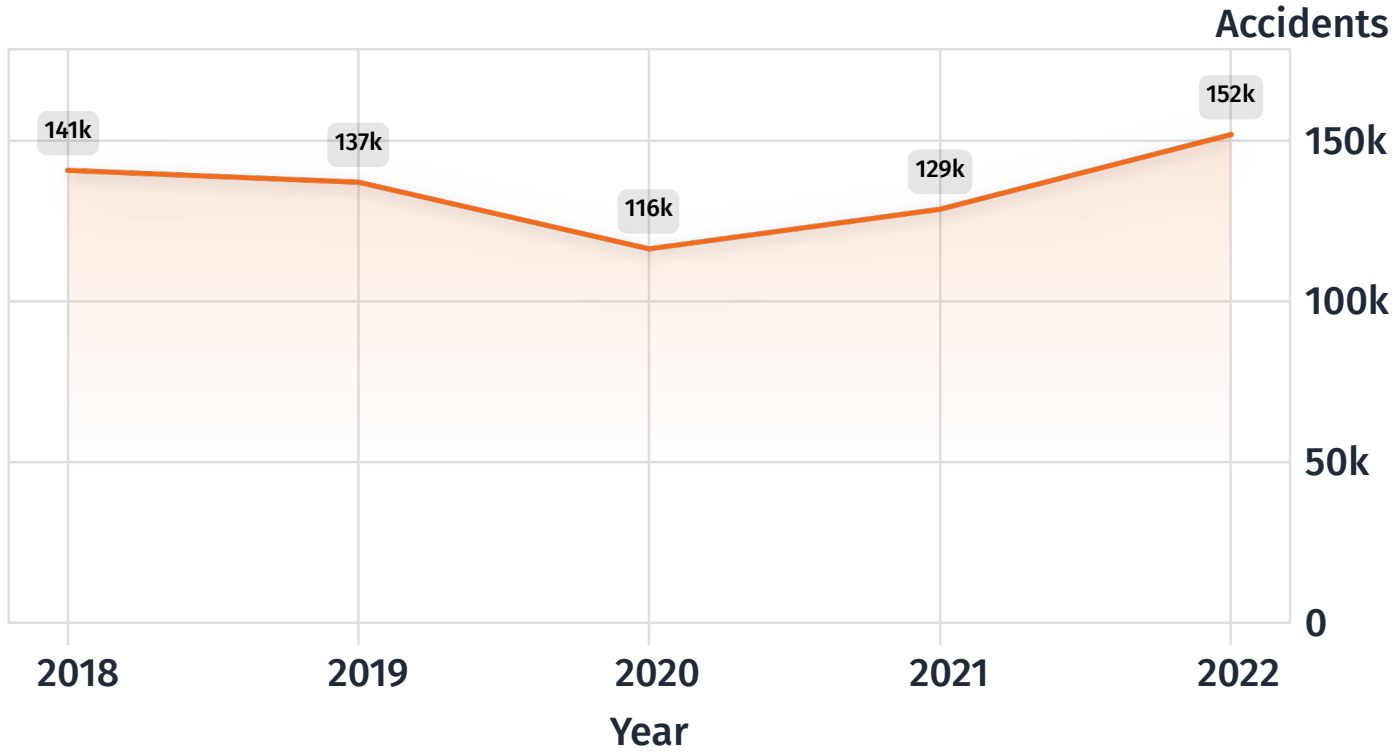
Null does not mean zero

Source: MoRTH / PIB

Made by Adarsh

NATIONAL HIGHWAYS

National Highway accidents rebounded sharply in 2022



2018: 140,843 | 2019: 137,191 | 2020: 116,496 | 2021: 128,825 | 2022: 151,997

2022 was 18.0% above 2021

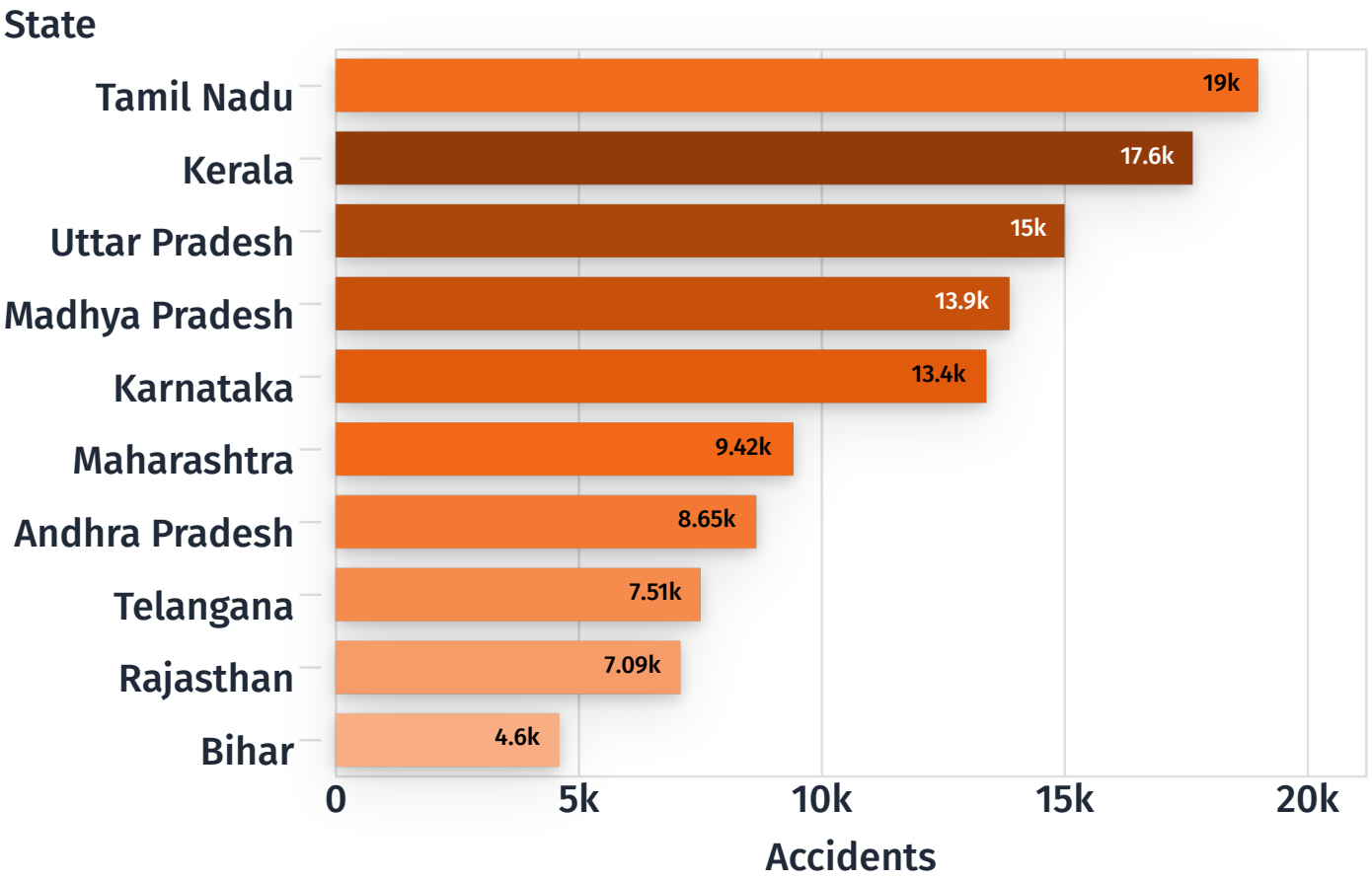
2022 was 7.9% above 2018

Source: MoRTH / PIB

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2022 TOP TEN

Tamil Nadu and Kerala lead the report's 2022 top-ten ranking



Ranked 2022 National Highway accidents: Tamil Nadu 18,972; Kerala 17,627; Uttar Pradesh 14,990; Madhya Pradesh 13,860; Karnataka 13,384; Maharashtra 9,417; Andhra Pradesh 8,650; Telangana 7,505; Rajasthan 7,093; Bihar 4,601.

Scope note: these are the report's top ten states, not all states.

Source: MoRTH / PIB

Made by Adarsh

SHARE OF NATIONAL TOTALS

National Highways carry a larger share of fatalities than accidents

32.94%

Accident share

36.22%

Fatality share

Source: MoRTH / PIB

Road group	Accidents	Fatalities
All roads	461,312	168,491
National Highways including Expressways	151,997	61,038

Takeaway: fatality share is higher than accident share. Counts are not exposure-adjusted risk rates.

Made by Adarsh

PROPOSED DASHBOARD LAYOUT

Use the dashboard to connect the workflow to three findings

Proposed dashboard layout — not a live Power BI screenshot.

2022 NH accidents	2022 NH fatalities	NH accident share	NH fatality share

NH trend line and ranked top-ten state bars

Year slicer; state filtering only where the data supports it

NH accidents rebounded in 2022

Tamil Nadu had the largest NH count among the report's top ten

NH fatality share exceeded its accident share

Future scope: add district/route-level e-DAR data, traffic volume, and highway length; add other variables only when sourced reliably. Scope note: state-level NH counts cover the report's top ten only.

Thank you

Source: MoRTH / PIB

Made by Adarsh